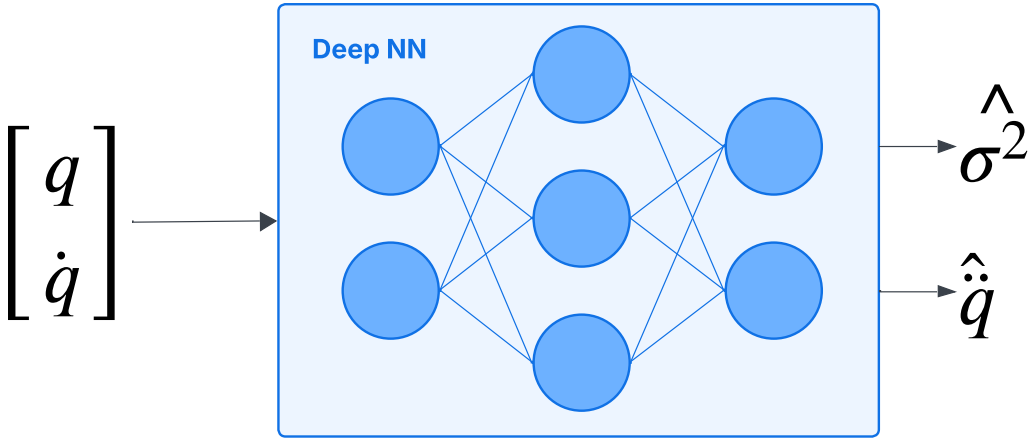
Deep NN
$\begin{bmatrix} q \\ \dot{q} \end{bmatrix}$
$\hat{\sigma^2}$
$\hat{\ddot{q}}$